SPATIAL ANALYSIS REPORT

India Regional Market Opportunity & Underserved Area Detection

Prepared from the supplied regional market dataset

Dataset: india_regional_market_v2.csv
Records analysed: 400
Geographic units: 10 city/state markets
Report date: 11 November 2026

Objective
Visualize regional market indicators geographically and identify three promising markets where demand potential is attractive relative to current competition and acquisition cost.

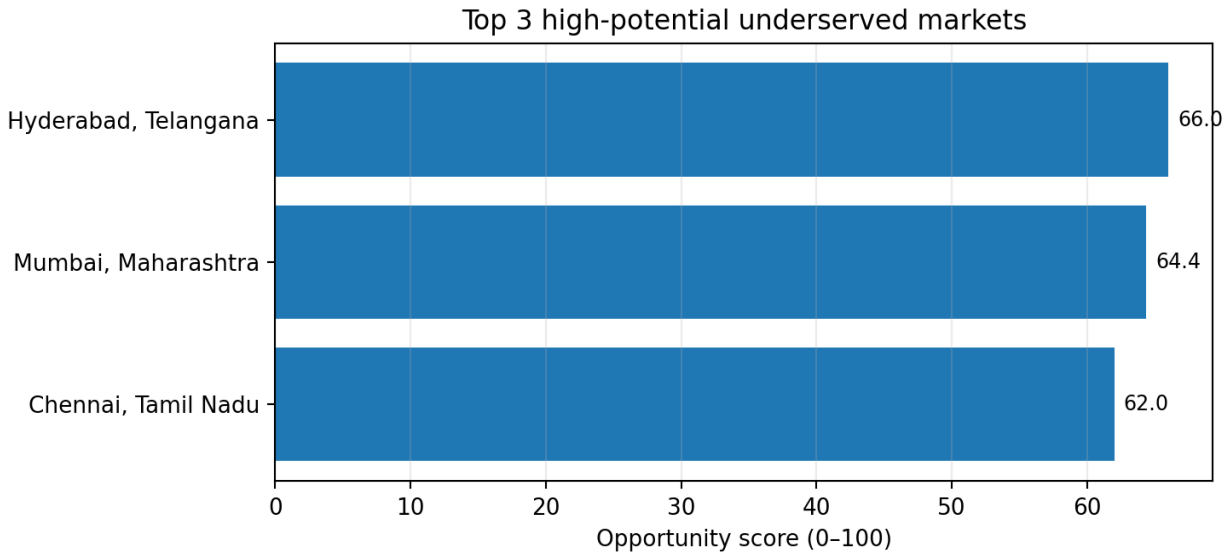
1. Executive Summary

The analysis cleaned and aggregated **400 valid regional records** into **10 city/state markets**. The opportunity ranking highlights **Hyderabad, Telangana**, **Mumbai, Maharashtra**, and **Chennai, Tamil Nadu** as the top three markets under the scoring method described in this report.

Important data limitation: The supplied CSV does not contain sales revenue or actual user-density fields. Therefore, revenue and user density could not be calculated directly. Monthly search demand is used as a demand proxy; population segment is retained for context, while income, lead conversion, competitor count and estimated acquisition cost are used as supporting market indicators. The results are screening priorities, not proof of unmet demand.

Top 3 recommended markets

Rank	City / State	Avg. monthly demand	Avg. competitors	Conversion	Avg. CAC (INR)	Score
1	Hyderabad, Telangana	1,145	7.2	8.2%	■439	66.0
2	Mumbai, Maharashtra	1,117	7.2	8.5%	■456	64.4
3	Chennai, Tamil Nadu	1,144	7.9	7.9%	■412	62.0



2. Dataset Cleaning & Preparation

The following preparation steps were applied before spatial aggregation:

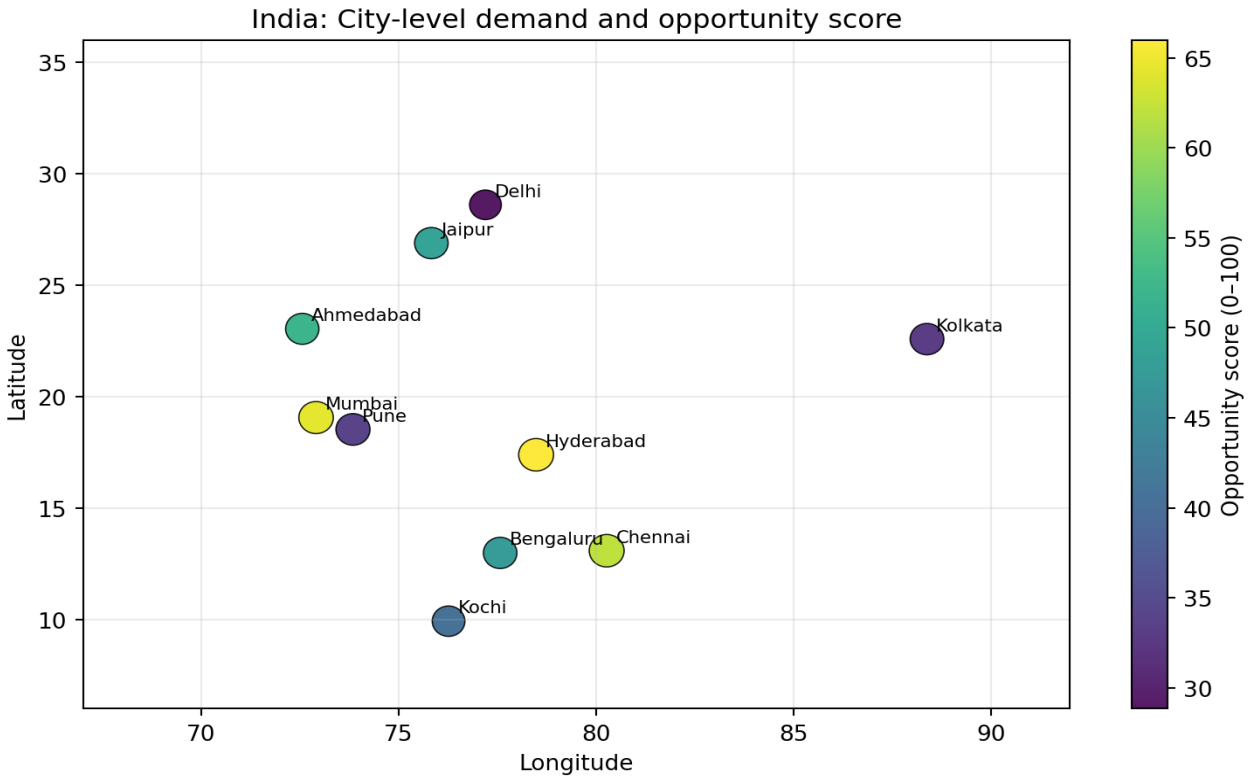
- Trimmed whitespace from text fields and converted numeric columns to numeric types.
- Removed duplicate rows and records missing required location or analysis fields.
- Checked latitude/longitude against a broad India coordinate range (latitude 6–38°, longitude 67–98°).
- Excluded negative monthly demand and competitor counts; constrained lead conversion rates to the 0–1 range.
- Grouped records by State and City and calculated mean market indicators for each city.

Data quality summary

Check	Result
Rows in source CSV	400
Rows retained after cleaning	400
Duplicate rows removed	0
City/state markets	10
Missing values in retained analysis fields	0

3. Geographic Visualization

The spatial scatter plot places each city at its average latitude and longitude. Bubble size represents average monthly search demand, while colour represents the calculated opportunity score. This is a point-based spatial visualization, not a state-boundary choropleth map.



Bubble size represents average monthly search demand. Points are city centroids from the dataset.

4. Ranking Method & Interpretation

Each city-level indicator was min–max normalized across the ten markets. The five normalized indicators were combined with equal weights:

Opportunity score = average of:

1. Higher average monthly search demand (positive factor)
2. Higher average lead conversion rate (positive factor)
3. Higher median annual income (positive factor)
4. Lower average existing competitor count (positive factor after reversal)
5. Lower estimated acquisition cost (positive factor after reversal)

The score ranges from 0 to 100 within this dataset. It is a relative ranking, not a probability or guaranteed return. Equal weighting is a transparent baseline; business teams may change weights based on strategy and validate them with real sales outcomes.

5. City-level Market Comparison

City	State	Records	Demand	Competitors	Income (■/yr)	Conversion	CAC (■)	Score
Hyderabad	Telangana	42	1145	7.2	665,610	8.2%	439	66.0
Mumbai	Maharashtra	28	1117	7.2	661,808	8.5%	456	64.4
Chennai	Tamil Nadu	47	1144	7.9	678,182	7.9%	412	62.0
Ahmedabad	Gujarat	50	1042	8.4	710,724	8.4%	487	52.0
Jaipur	Rajasthan	33	1056	7.7	668,222	8.0%	447	48.8
Bengaluru	Karnataka	32	1056	7.7	646,763	8.8%	476	47.4
Kochi	Kerala	39	1000	6.8	660,594	8.1%	520	40.5
Pune	Maharashtra	41	1087	7.9	665,735	7.2%	479	33.7
Kolkata	West Bengal	32	1066	8.6	649,084	7.3%	392	33.0
Delhi	Delhi	56	947	8.0	701,424	7.3%	486	28.8

6. Recommendations & Conclusion

1. Hyderabad, Telangana: First-priority market under the combined score. It has the strongest overall balance of demand, conversion, income, competition and acquisition-cost indicators in this dataset.

2. Mumbai, Maharashtra: Second-priority market. Its relatively strong demand and low average competitor count support further validation, although acquisition cost should be monitored.

3. Chennai, Tamil Nadu: Third-priority market. It shows the highest average demand among the listed city markets and a competitive overall score; local competition and customer segments should be checked before expansion.

Recommended next steps

• Validate the top three markets using actual sales revenue, active customers, conversion funnel and customer-density data.
• Compare demand by smaller postal-code or neighbourhood units before selecting a launch location.
• Verify competitor locations and service coverage to confirm whether a market is genuinely underserved.
• Run a pilot campaign and compare customer acquisition cost and conversion against the dataset estimates.
• Refresh the analysis regularly as demand and competitor conditions change.

Conclusion

The dataset supports a preliminary geographic opportunity ranking, with Hyderabad, Mumbai and Chennai emerging as the top three city markets. The spatial scatter visualization helps compare location, demand and relative opportunity. Since revenue and user-density fields are absent, the findings should be treated as an initial screening report and validated with operational data before making expansion decisions.

Appendix: Dataset fields used

RegionID, City, State, Latitude, Longitude, PopulationSegment, MonthlySearchDemand, ExistingCompetitors, MedianAnnualIncomeINR, LeadConversionRate and EstimatedAcquisitionCostINR.

Prepared as a spatial analysis deliverable based on the supplied CSV and the task guidelines shown in the reference image.